The finite J-norm is dominated by the discrete K-norm

(the direction $\mathcal{J} \lesssim \|\cdot\|_K$ of Janson's Lemma 1, over a monoid)

1 Setting and statement

Let α be an additive commutative monoid with a preorder $\le$ and a map $x \mapsto |x|$ from α to itself satisfying

$$|x+y| \le |x| + |y| \qquad (x, y \in \alpha), \tag{1}$$

together with the following *refined Riesz decomposition*:

(R) if $|x| \le |y| + |z|$, then there are $x_0, x_1 \in \alpha$ with

$$x = x_0 + x_1, \qquad |x_0| \le |y|, \quad |x_1| \le |z|, \qquad |x_0| \le |x|, \quad |x_1| \le |x|.$$

The first three conditions of (R) are the field `exists_decomp` of `Abs.IsModulus`; the last two are new. They hold for the three existing instances with the witnesses already used there: on $\mathbb{R}$, x_0 is x clamped to $[-|y|, |y|]$ and $x_1 = x - x_0$ has the sign of x; on $[0,\infty]$, $x_0 = \min(x, y)$ and $x_1 = x - x_0$; on products, (R) is checked pointwise.

Let A_0, A_1 be quasinorms on α with values in $[0,\infty]$ that are *solid*: $|x| \le |y|$ implies $\|x\|_i \le \|y\|_i$, where $\|\cdot\|_i$ is the quasinorm of A_i. No other property of A_0, A_1 is used; in particular their quasi-triangle constants never appear.

Fix $0 < \theta < 1$ and $0 < q \le \infty$. For $\nu \in \mathbb{Z}$, $x, u \in \alpha$ put

$$K_\nu(x) = \inf_{x = x_0 + x_1} \big(\|x_0\|_0 + 2^\nu \|x_1\|_1\big), \qquad J_\nu(u) = \max\big(\|u\|_0,\ 2^\nu \|u\|_1\big),$$

and $\Delta = \{x \in \alpha : \|x\|_0 < \infty,\ \|x\|_1 < \infty\}$. For a family $f = (f_\nu)_{\nu \in S}$ in $[0,\infty]$ indexed by a countable set S, $\|f\|_{\ell^q}$ is $(\sum_{\nu \in S} f_\nu^q)^{1/q}$ if $q < \infty$ and $\sup_{\nu \in S} f_\nu$ if $q = \infty$. The two norms to be compared are

$$\|x\|_K = \big\|(2^{-\nu\theta} K_\nu(x))_{\nu \in \mathbb{Z}}\big\|_{\ell^q} \qquad (\texttt{discreteKMethod}),$$

$$\mathcal{J}(x) = \inf\Big\{ \big\|(2^{-\nu\theta} J_\nu(u_\nu))_{\nu \in \mathbb{Z}}\big\|_{\ell^q} : u \colon \mathbb{Z} \to \alpha \text{ finitely supported, } \textstyle\sum_{\nu \in \mathbb{Z}} u_\nu = x \Big\} \qquad (\texttt{jInfNorm}).$$

Since $J_\nu(0) = 0$, a family $u \colon \{-N, \dots, N-1\} \to \alpha$ with $\sum_{\nu=-N}^{N-1} u_\nu = x$, extended by zero to $\mathbb{Z}$, is admissible, with cost $\|(2^{-\nu\theta} J_\nu(u_\nu))_{\nu=-N}^{N-1}\|_{\ell^q}$. Conversely every admissible u is supported in $\{-N, \dots, N-1\}$ for some N. Hence $\mathcal{J}$ is equivalently the infimum of these costs over $N \in \mathbb{N}$ and $u \colon \{-N, \dots, N-1\} \to \alpha$.

Theorem 1. *For every $x \in \Delta$,*

$$\mathcal{J}(x) \le 2^{\min(\theta, 1-\theta)}\, \|x\|_K.$$

The constant is at most $\sqrt{2}$ and does not depend on q or on the couple.

Idea of the proof. For each level μ choose a near-optimal decomposition $x = y_\mu + z_\mu$ for $K_\mu(x)$. Using (R), split x into $2N$ pieces, the piece u_ν being dominated by $y_{\nu+1}$ and by z_ν. Then $J_\nu(u_\nu) \lesssim K_{\nu+1}(x)$, except for one of the two norms of each of the two end pieces, which is controlled only by x itself. After weighting, these end terms decay geometrically in N, because $0 < \theta < 1$ and $x \in \Delta$. The ℓ^q-norm is estimated through its r-th power, $r = \min(q, 1)$, which is subadditive with constant 1. The error terms vanish in the limit, so the main term keeps the constant 2^θ for every q.

2 Elementary facts

Lemma 2. *For every $x \in \alpha$ and $\nu \in \mathbb{Z}$, $K_\nu(x) \le K_{\nu+1}(x) \le 2K_\nu(x)$.*

Proof. For any decomposition $x = x_0 + x_1$,

$$\|x_0\|_0 + 2^\nu\|x_1\|_1 \le \|x_0\|_0 + 2^{\nu+1}\|x_1\|_1 \le 2\big(\|x_0\|_0 + 2^\nu\|x_1\|_1\big).$$

Taking the infimum over all decompositions gives both inequalities. □

No finiteness assumption on x is needed; the hypothesis of `monotone_kNorm` in `Basic.lean` is superfluous for dyadic points. The second inequality is a case of `kNorm_le_mul`. The main proof uses only the first one.

Lemma 3. *For every $x \in \alpha$, $\mu \in \mathbb{Z}$ and $\delta > 0$ there is a decomposition $x = y + z$ with $\|y\|_0 + 2^\mu\|z\|_1 \le K_\mu(x) + \delta$.*

Proof. If $K_\mu(x) = \infty$, any decomposition works, e.g. $x = 0 + x$. Otherwise $K_\mu(x) < K_\mu(x) + \delta$, and by the definition of the infimum some decomposition has cost below $K_\mu(x)+\delta$ (`exists_decomp_lt_of_lt_kNor`
□

An additive error is used because the infimum need not be attained, and a multiplicative error $1 + \varepsilon$ is useless when $K_\mu(x) = 0$.

Lemma 4. *Put $r = \min(q, 1)$. For $f, g\colon \mathbb{Z} \to [0, \infty]$,*

$$\|f + g\|_{\ell^q}^r \le \|f\|_{\ell^q}^r + \|g\|_{\ell^q}^r.$$

Proof. If $q \ge 1$, including $q = \infty$, then $r = 1$ and this is Minkowski's inequality. Let $0 < q < 1$, so that $r = q$ and $\|f\|_{\ell^q}^q = \sum_{\nu\in\mathbb{Z}} f_\nu^q$. For $s, t \in [0, \infty]$ we have $(s + t)^q \le s^q + t^q$. This is clear if $s + t$ is 0 or ∞; otherwise it follows from $\lambda^q \ge \lambda$ for $\lambda \in [0, 1]$, applied to $\lambda = s/(s + t)$ and $\lambda = t/(s + t)$. Summing this inequality with $s = f_\nu$, $t = g_\nu$ over $\nu \in \mathbb{Z}$ gives the claim. □

In Mathlib, Lemma 4 is `inhmgELpNorm_add_le`. For $q \ne 0$, $\|f\|_{\ell^q}^r$ equals `inhmgELpNorm f q Measure.count`, by

- `inhmgELpNorm_eq_eLpNorm` if $q \ge 1$,
- `inhmgELpNorm_eq_eLpNorm_pow` if $q \le 1$.

Lemma 5. *Let $f, g\colon \mathbb{Z} \to [0, \infty]$.*

(a) If $c \in [0, \infty)$ and $f_\nu \le c\, g_\nu$ for all $\nu \in \mathbb{Z}$, then $\|f\|_{\ell^q} \le c\|g\|_{\ell^q}$.

(b) For $k \in \mathbb{Z}$, $\|(f_{\nu+k})_{\nu\in\mathbb{Z}}\|_{\ell^q} = \|f\|_{\ell^q}$.

(c) If $S \subseteq \mathbb{Z}$ is finite with n elements and $c \in [0,\infty]$, then $\|c\,\mathbf{1}_S\|_{\ell^q} \le c\,n^{1/q}$, where $1/\infty = 0$.

Proof. (a) The ℓ^q-norm is monotone and positively homogeneous. (b) The map $\nu \mapsto \nu + k$ is a bijection of $\mathbb{Z}$, and sums and suprema are invariant under reindexing. (c) If $q < \infty$, then $\|c\,\mathbf{1}_S\|_{\ell^q}^q = nc^q$. If $q = \infty$, then $\|c\,\mathbf{1}_S\|_{\ell^\infty}$ is c or 0, hence at most $c = c\,n^0$. □

In Lean:

(a) `eLpNorm_le_mul_eLpNorm_of_ae_le_mul'` with $c : \mathbb{R}_{\ge 0}$;

(b) `eLpNorm_comp_measurePreserving` with `measurePreserving_add_right Measure.count k`;

(c) `eLpNorm_indicator_const_le` with `Measure.count_apply_finset`.

Every function on $\mathbb{Z}$ is measurable (`measurable_of_countable`).

3 Splitting x into $2N$ pieces

This is the only place where (R) is used. The first piece is split off r, and the rest of r is split by induction, so no subtraction is needed.

Proposition 6. *Let $x \in \alpha$, $a \in \mathbb{Z}$, $m \in \mathbb{N}$, put $I = \{a, \dots, a+m\}$, and for $\mu = a+1, \dots, a+m$ let $x = y_\mu + z_\mu$ be arbitrary decompositions. Let $r \in \alpha$ with $|r| \le |x|$. Then there is $u\colon \mathbb{Z} \to \alpha$ with $u_\nu = 0$ for $\nu \notin I$, $\sum_{\nu\in\mathbb{Z}} u_\nu = r$ and, for every $\nu \in I$,*

(a) $|u_\nu| \le |y_{\nu+1}|$ if $\nu < a+m$;

(b) $|u_\nu| \le |z_\nu|$ if $\nu > a$;

(c) $|u_\nu| \le |r|$.

Proof. By induction on m, for all a and r simultaneously.

The case $m = 0$. Let $u_a = r$ and $u_\nu = 0$ for $\nu \ne a$. Then (a) and (b) are vacuous, and (c) holds by reflexivity.

The step from m to $m+1$. Since $x = y_{a+1} + z_{a+1}$, inequality (1) and transitivity give

$$|r| \le |x| \le |y_{a+1}| + |z_{a+1}|.$$

Applying (R) to r gives $r = v + r'$ with

$$|v| \le |y_{a+1}|, \qquad |r'| \le |z_{a+1}|, \qquad |v| \le |r|, \qquad |r'| \le |r|. \tag{2}$$

By the last inequality, $|r'| \le |x|$, so the induction hypothesis applies to the start $a+1$, the length m and the remainder r'. It gives $u'\colon \mathbb{Z} \to \alpha$ vanishing outside $I' = \{a+1, \dots, a+1+m\}$ with $\sum_{\nu\in\mathbb{Z}} u'_\nu = r'$ and (a)–(c) relative to I' and r'. Define $u_a = v$ and $u_\nu = u'_\nu$ for $\nu \ne a$. Then u vanishes outside $I = \{a\} \cup I'$, and since $u'_a = 0$, $\sum_{\nu\in\mathbb{Z}} u_\nu = v + \sum_{\nu\in\mathbb{Z}} u'_\nu = v + r' = r$.

At $\nu = a$, (a) and (c) are the first and third inequality of (2), and (b) is vacuous. Let $\nu \in I'$, so $u_\nu = u'_\nu$. Property (a) is that of u', since the right end of I' is that of I. Property (c) follows from $|u'_\nu| \le |r'| \le |r|$. For (b), if $\nu > a+1$ it is that of u'. At $\nu = a+1$ the induction hypothesis gives no condition (b), because $a+1$ is the left end of I'; instead, (c) for u' and the second inequality of (2) give $|u'_{a+1}| \le |r'| \le |z_{a+1}|$. □

Both new conditions of (R) are needed. The condition $|x_1| \le |x|$ gives $|r'| \le |x|$, without which the induction hypothesis could not be applied; the condition $|x_0| \le |x|$, through (c), gives (b) at the left end of I'. For the same reason (c) is stated relative to r and not to x.

Proposition 7. *Let $N \ge 1$, $x \in \alpha$, and for $\mu = -N+1, \dots, N-1$ let $x = y_\mu + z_\mu$ be arbitrary decompositions. Then there is $u\colon \mathbb{Z} \to \alpha$ vanishing outside $\{-N, \dots, N-1\}$ with $\sum_{\nu\in\mathbb{Z}} u_\nu = x$ and, for every $\nu \in \{-N, \dots, N-1\}$,*

(a) $|u_\nu| \le |y_{\nu+1}|$ if $\nu \le N-2$;

(b) $|u_\nu| \le |z_\nu|$ if $\nu \ge -N+1$;

(c) $|u_\nu| \le |x|$.

Proof. Apply Proposition 6 with $a = -N$, $m = 2N-1$ and $r = x$; then $I = \{-N, \dots, N-1\}$. □

4 The pointwise estimate

Lemma 8. *Let $x \in \alpha$, $\delta > 0$ and $N \ge 1$. For $\mu = -N+1, \dots, N-1$ choose decompositions $x = y_\mu + z_\mu$ as in Lemma 3, i.e.*

$$\|y_\mu\|_0 + 2^\mu \|z_\mu\|_1 \le K_\mu(x) + \delta, \tag{3}$$

and let $u\colon \mathbb{Z} \to \alpha$ be given by Proposition 7. Put $E_{-N} = 2^{-N}\|x\|_1$, $E_{N-1} = \|x\|_0$ and $E_\nu = 0$ for the other $\nu \in \{-N, \dots, N-1\}$. Then for every $\nu \in \{-N, \dots, N-1\}$,

$$J_\nu(u_\nu) \le \max\big(K_{\nu+1}(x) + \delta,\ E_\nu\big).$$

Proof. Since $N \ge 1$, the indices $-N$ and $N-1$ are distinct, so each ν is exceptional in at most one of the two entries of $J_\nu(u_\nu)$. We bound both entries by the right-hand side; solidity turns each modulus inequality of Proposition 7 into an inequality of quasinorms.

The entry $\|u_\nu\|_0$. If $\nu \le N-2$, then by (a) and (3) at $\mu = \nu+1$, $\|u_\nu\|_0 \le \|y_{\nu+1}\|_0 \le K_{\nu+1}(x) + \delta$. If $\nu = N-1$, then by (c), $\|u_\nu\|_0 \le \|x\|_0 = E_{N-1}$.

The entry $2^\nu\|u_\nu\|_1$. If $\nu \ge -N+1$, then by (b), (3) at $\mu = \nu$, and Lemma 2, $2^\nu\|u_\nu\|_1 \le 2^\nu\|z_\nu\|_1 \le K_\nu(x) + \delta \le K_{\nu+1}(x) + \delta$. If $\nu = -N$, then by (c), $2^{-N}\|u_\nu\|_1 \le 2^{-N}\|x\|_1 = E_{-N}$. □

5 Proof of Theorem 1

We prove $\mathcal{J}(x) \le 2^\theta \|x\|_K$; the bound with $2^{1-\theta}$ is Remark 10. Fix $x \in \Delta$, and put $r = \min(q, 1)$ and $a_\mu = 2^{-\mu\theta} K_\mu(x)$ for $\mu \in \mathbb{Z}$, so that $\|a\|_{\ell^q} = \|x\|_K$. Fix $N \ge 1$ and $\delta > 0$, let u be as in Lemma 8, and put

$$b_\nu = 2^{-\nu\theta} J_\nu(u_\nu) \quad (\nu \in \mathbb{Z}), \qquad e_\nu = 2^{-\nu\theta} E_\nu \quad (\nu \in \{-N, \dots, N-1\}),$$

and $e_\nu = 0$ for the other $\nu \in \mathbb{Z}$. The end terms are

$$e_{-N} = 2^{N\theta} 2^{-N} \|x\|_1 = 2^{-(1-\theta)N} \|x\|_1, \qquad e_{N-1} = 2^{-(N-1)\theta} \|x\|_0.$$

Since u vanishes outside $\{-N, \dots, N-1\}$ and $\sum_{\nu\in\mathbb{Z}} u_\nu = x$, the family u is admissible in the definition of $\mathcal{J}$, so $\mathcal{J}(x) \le \|b\|_{\ell^q}$.

Lemma 9. *For every $\nu \in \mathbb{Z}$,*

$$b_\nu \le 2^\theta a_{\nu+1} + \delta\, 2^{N\theta}\, \mathbf{1}_{\{-N,\dots,N-1\}}(\nu) + e_\nu.$$

Proof. Let $\nu \in \{-N,\dots,N-1\}$. Bounding the maximum in Lemma 8 by the sum gives $J_\nu(u_\nu) \le K_{\nu+1}(x) + \delta + E_\nu$, and we multiply this by $2^{-\nu\theta}$. Since $2^{-\nu\theta} = 2^\theta\, 2^{-(\nu+1)\theta}$, the first term becomes $2^\theta a_{\nu+1}$. Since $\nu \ge -N$ and $\theta > 0$, we have $2^{-\nu\theta} \le 2^{N\theta}$, which bounds the second term. The third term becomes e_ν.

Let $\nu \notin \{-N,\dots,N-1\}$. Then $u_\nu = 0$, so $b_\nu = 2^{-\nu\theta} J_\nu(0) = 0$, because quasinorms vanish at 0. □

The main term carries no indicator, so that Lemma 5(b) applies to it.

Since $N \ge 1$, the indices $-N$ and $N-1$ are distinct, so $e = e_{-N}\mathbf{1}_{\{-N\}} + e_{N-1}\mathbf{1}_{\{N-1\}}$. By Lemma 9 and Lemma 5(a) with $c = 1$, $\|b\|_{\ell^q}$ is at most the ℓ^q-norm of the sum of the four families

$$(2^\theta a_{\nu+1})_{\nu\in\mathbb{Z}}, \qquad \delta\, 2^{N\theta}\, \mathbf{1}_{\{-N,\dots,N-1\}}, \qquad e_{-N}\mathbf{1}_{\{-N\}}, \qquad e_{N-1}\mathbf{1}_{\{N-1\}}.$$

By Lemma 4, applied three times, $\|b\|_{\ell^q}^r$ is at most the sum of the r-th powers of their norms. By Lemma 5(a) and (b) with $k = 1$, the norm of the first family is at most $2^\theta \|a\|_{\ell^q} = 2^\theta \|x\|_K$. By Lemma 5(c), the norms of the other three are at most $\delta\, 2^{N\theta}(2N)^{1/q}$, e_{-N} and e_{N-1}. Hence

$$\mathcal{J}(x)^r \le 2^{\theta r}\|x\|_K^r + \left(\delta\, 2^{N\theta}(2N)^{1/q}\right)^r + e_{-N}^r + e_{N-1}^r. \tag{4}$$

The left-hand side of (4) depends on neither δ nor N. Fix N and let $\delta \to 0$. Since $2^{N\theta}(2N)^{1/q}$ is finite and $r > 0$, the second term tends to 0, and since addition on $[0,\infty]$ is continuous,

$$\mathcal{J}(x)^r \le 2^{\theta r}\|x\|_K^r + e_{-N}^r + e_{N-1}^r.$$

Now let $N \to \infty$. Since $\theta < 1$ and $\|x\|_1 < \infty$, $e_{-N} \to 0$; since $\theta > 0$ and $\|x\|_0 < \infty$, $e_{N-1} \to 0$. As $r > 0$, their r-th powers tend to 0 as well. This gives $\mathcal{J}(x)^r \le 2^{\theta r}\|x\|_K^r$, and taking r-th roots proves the claim. □

The two limits are taken one after the other: $\delta \to 0$ for fixed N, then $N \to \infty$. This is legitimate because $\mathcal{J}(x)$ depends on neither, and the end terms do not depend on δ. No assumption $\|x\|_K < \infty$ is needed, and no case distinction between $q < \infty$ and $q = \infty$. In Lean both limits are taken with `ge_of_tendsto'`, the first along $\delta = (n+1)^{-1}$, $n \to \infty$.

6 Remarks

Remark 10 (The constant $2^{1-\theta}$). Feed Proposition 7 the decompositions one level lower: for $\mu = -N+1,\dots,N-1$, let $x = y_\mu + z_\mu$ satisfy (3) at level $\mu-1$. For $\nu \le N-2$, (a) and Lemma 2 give $\|u_\nu\|_0 \le K_\nu(x) + \delta \le 2(K_{\nu-1}(x) + \delta)$; for $\nu \ge -N+1$, (b) gives $2^\nu\|u_\nu\|_1 = 2\cdot 2^{\nu-1}\|u_\nu\|_1 \le 2(K_{\nu-1}(x)+\delta)$. The exceptional entries are unchanged, so $J_\nu(u_\nu) \le \max(2(K_{\nu-1}(x)+\delta), E_\nu)$. Since $2\cdot 2^{-\nu\theta} = 2^{1-\theta}2^{-(\nu-1)\theta}$, the rest of the proof goes through with $2^\theta a_{\nu+1}$ replaced by $2^{1-\theta}a_{\nu-1}$, δ by 2δ, and $k = 1$ by $k = -1$ in Lemma 5(b). This uses the second inequality of Lemma 2. Choosing the better of the two versions gives the constant $2^{\min(\theta,1-\theta)}$.

Remark 11 (Where the hypotheses are used).

- (R), (1) and transitivity of $\le$: only in Proposition 6.

- Solidity of A_0, A_1: only in Lemma 8.
- $x \in \Delta$ and $0 < \theta < 1$: in $e_{-N}, e_{N-1} \to 0$. Each end uses one half of each condition: e_{-N} uses $\theta < 1$ and $\|x\|_1 < \infty$, e_{N-1} uses $\theta > 0$ and $\|x\|_0 < \infty$. Besides, $\theta > 0$ gives the bound $2^{-\nu\theta} \leq 2^{N\theta}$ in Lemma 9, where any finite bound would do.
- The quasi-triangle inequalities of A_0 and A_1: never. For ℓ^q, only Lemma 4 is used. Its constant is 1 for every q, which is why the constant of Theorem 1 does not depend on q; the quasi-triangle inequality of ℓ^q would contribute a factor $2^{1/q-1}$ for $q < 1$.

For $q = 0$, Mathlib's `eLpNorm` is identically 0. The family with $u_0 = x$ and $u_\nu = 0$ for $\nu \neq 0$ (`Pi.single 0 x`) is admissible, so $\mathcal{J}(x) = 0$ and the inequality is trivial. This case is excluded from the proof above because there $r = 0$.

Remark 12 (Why $x \in \Delta$ is necessary). Let $q \neq 0$. If $\mathcal{J}(x) < \infty$, some admissible ℓ^q-norm is finite, so each $J_\nu(u_\nu)$ is finite and each u_ν lies in Δ. Since Δ is an additive submonoid (by the quasi-triangle inequality with finite constants), the finite sum x lies in Δ. Hence $\mathcal{J} = \infty$ outside Δ, whereas $\|x\|_K$ may be finite there.